# SPAZA STOCK & SALES TRACKER

***Complete Project Documentation & Samsung Innovation Internship 2027 Submission***

***Author: Romeo Mangena | Establishment Study: Dlamini General Spaza Shop (Soweto) | Date: November 2026***

## 1. Executive Summary & Project Overview

The Spaza Stock & Sales Tracker is a full-stack web application designed to help spaza shops and small township retailers manage inventory, record point-of-sale (POS) transactions, monitor stock levels in real time, and generate smart restocking recommendations. The system eliminates manual notebook record-keeping, prevents stockouts, and provides vital business insights.

## 2. Problem Statement & Route A Establishment Study

Many small retailers in townships and local communities still rely on paper notebooks and memory to track stock and sales. Field studies conducted at Dlamini General Spaza Shop in Soweto revealed that manual record-keeping leads to inaccurate inventory counts, missing sales records, and unexpected stockouts of fast-moving staples (such as milk, bread, and maize meal), costing the business valuable revenue and customers.

## 3. Solution Architecture & Tech Stack

Backend: Spring Boot 3 (Java 21) REST API featuring Spring Security, JWT authentication, Spring Data JPA, and H2/MySQL database persistence. Frontend: Single-Page Application built with React 18, Tailwind CSS, and Lucide UI components, served directly from Spring Boot static resources for seamless standalone deployment.

## 4. Core Features & Smart Restocking Engine

- Real-Time Inventory Management: Full CRUD operations with instant search and low-stock flagging.

- Point-of-Sale (POS) Checkout: Over-the-counter transaction recording with automatic inventory decrement and stock validation.

- Smart Restocking Recommendations: Automated reorder engine calculating recommended order quantities using the formula: Recommended Quantity = (Reorder Level x 2) - Current Stock, complete with estimated supplier costs.

- Role-Based Access Control: Dedicated roles for OWNER (full management, reports, restocks) and STAFF (inventory lookup, sales recording).

## 5. Evaluator Test Access Credentials

To satisfy Samsung Innovation Internship evaluation requirements, test accounts are auto-seeded at startup:

â€¢ Owner Account: username 'owner' | password 'password123' (Full managerial access)

â€¢ Staff Account: username 'staff' | password 'password123' (Cashier POS access)

## 6. Database Schema & API Endpoints

- users table: id, username, password, full_name, role, created_at

- products table: id, name, category, price, quantity, reorder_level, barcode, supplier_name

- sales table: id, product_id, quantity, unit_price, total_price, sale_date, recorded_by_username

- REST Endpoints: /api/auth/**, /api/products/**, /api/sales/**, /api/dashboard/summary

## 7. Future Enhancements

- Barcode scanning support using mobile/USB scanners.

- Automated SMS notifications to local wholesale suppliers when low-stock thresholds are breached.

- AI-powered demand forecasting (LiteRT) based on community payday cycles and historical sales trends.